

Comparative Pangenomic Analysis of *Limosilactobacillus fermentum*

Comparative genomic analysis of the *Limosilactobacillus fermentum* isolates was performed using BRIG, with the reference genome comprising 2,020,722 bp (Figure 1). The concentric BLAST rings revealed a high degree of sequence conservation across the genomes, as evidenced by the extensive continuous regions of high sequence identity distributed throughout the chromosome. Most isolates exhibited substantial genomic coverage relative to the reference, indicating a conserved core genomic backbone within the species.

Despite overall conservation, several regions showed reduced sequence similarity or were absent in one or more isolates, appearing as discontinuities and gaps within the BLAST rings. These variable regions were distributed across multiple chromosomal loci and were not uniformly shared among all strains, suggesting accessory genomic elements and strain-specific genetic content. The observed variability was more pronounced in certain isolates, reflecting varying degrees of genomic divergence from the reference genome.

Analysis of GC content and GC skew demonstrated a generally stable genomic composition across the chromosome, although localized fluctuations were evident in several regions. These compositional variations coincided with some of the variable genomic loci identified in the comparative analysis, suggesting potential hotspots of genomic plasticity. Collectively, the BRIG analysis indicates that while *L. fermentum* genomes share extensive sequence conservation, substantial accessory genome variation exists among strains, contributing to intraspecies genomic diversity.

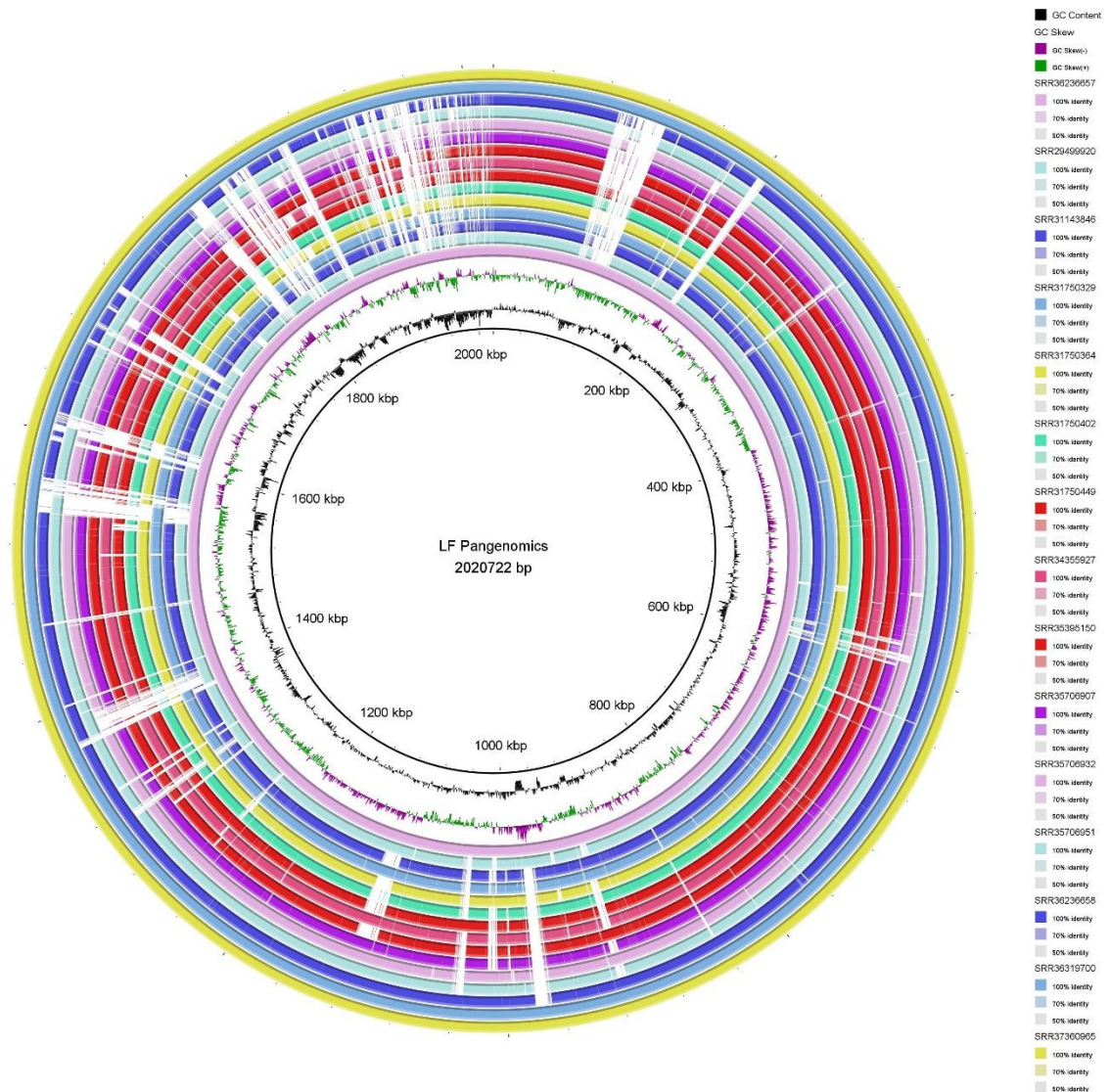


Figure 1. Comparative pangenomic visualization of *Limosilactobacillus fermentum* genomes generated using BRIG.

The circular BLAST Ring Image Generator (BRIG) plot compares multiple *Limosilactobacillus fermentum* genomes against a reference genome of 2,020,722 bp located at the center of the figure. Concentric rings represent individual isolates, with color intensity corresponding to sequence similarity thresholds of 100%, 70%, and 50% identity relative to the reference genome. Continuous colored regions indicate conserved genomic segments, whereas white gaps denote regions of reduced similarity or absence in the corresponding isolate. The innermost tracks

represent GC content (black) and GC skew (magenta and green), illustrating compositional variation across the chromosome. The extensive conservation observed across most genomic regions indicates a shared core genome, while discontinuous regions highlight accessory genomic loci and strain-specific genetic variation among the analyzed isolates.

The comparative BRIG analysis demonstrated that the *Limosilactobacillus fermentum* isolates possess a highly conserved genomic backbone, supporting the existence of a substantial core genome shared among strains. This high level of conservation is consistent with species-level genomic stability and suggests preservation of essential genes required for fundamental cellular processes and adaptation to common ecological niches. Nevertheless, the presence of multiple variable regions across the chromosome indicates considerable accessory genome diversity within the species. Such genomic variability may arise through horizontal gene transfer, gene gain and loss events, prophage acquisition, or the integration of mobile genetic elements, all of which contribute to bacterial genome evolution and ecological adaptation. The localized GC-content deviations observed in some variable regions further support the possibility of horizontally acquired DNA. These accessory genomic regions may encode functions associated with niche adaptation, carbohydrate utilization, stress tolerance, antimicrobial activity, host interaction, or other strain-specific phenotypes. Consequently, the observed genomic diversity among *L. fermentum* isolates likely reflects evolutionary adaptation to distinct environmental and host-associated habitats while maintaining a conserved core set of essential genes.

Comparative Genomic Analysis of *Limosilactobacillus fermentum* Using a Reference Genome

Comparative genome analysis was performed using BRIG with a reference *Limosilactobacillus fermentum* genome of 1,905,587 bp. The resulting circular comparison revealed extensive sequence conservation among the analyzed isolates, with most genomes exhibiting near-complete coverage of the reference chromosome. High levels of nucleotide identity were observed across the majority of genomic regions, indicating substantial conservation of the species' core genomic framework.

Despite the overall genomic similarity, several localized regions displayed reduced sequence identity or absence in one or more isolates, appearing as discontinuities within the BLAST rings. These variable regions were distributed across the chromosome and likely represent accessory genomic elements, including strain-specific genes and mobile genetic components. However, the limited extent of these discontinuities relative to the total genome size suggests that genomic divergence among the isolates is restricted to specific loci rather than being genome-wide.

The GC-content and GC-skew tracks exhibited characteristic bacterial chromosomal patterns, with generally stable nucleotide composition across the genome and localized compositional deviations at selected regions. These observations indicate a largely conserved genomic architecture with discrete regions of genomic plasticity contributing to strain-level diversity within *L. fermentum*.

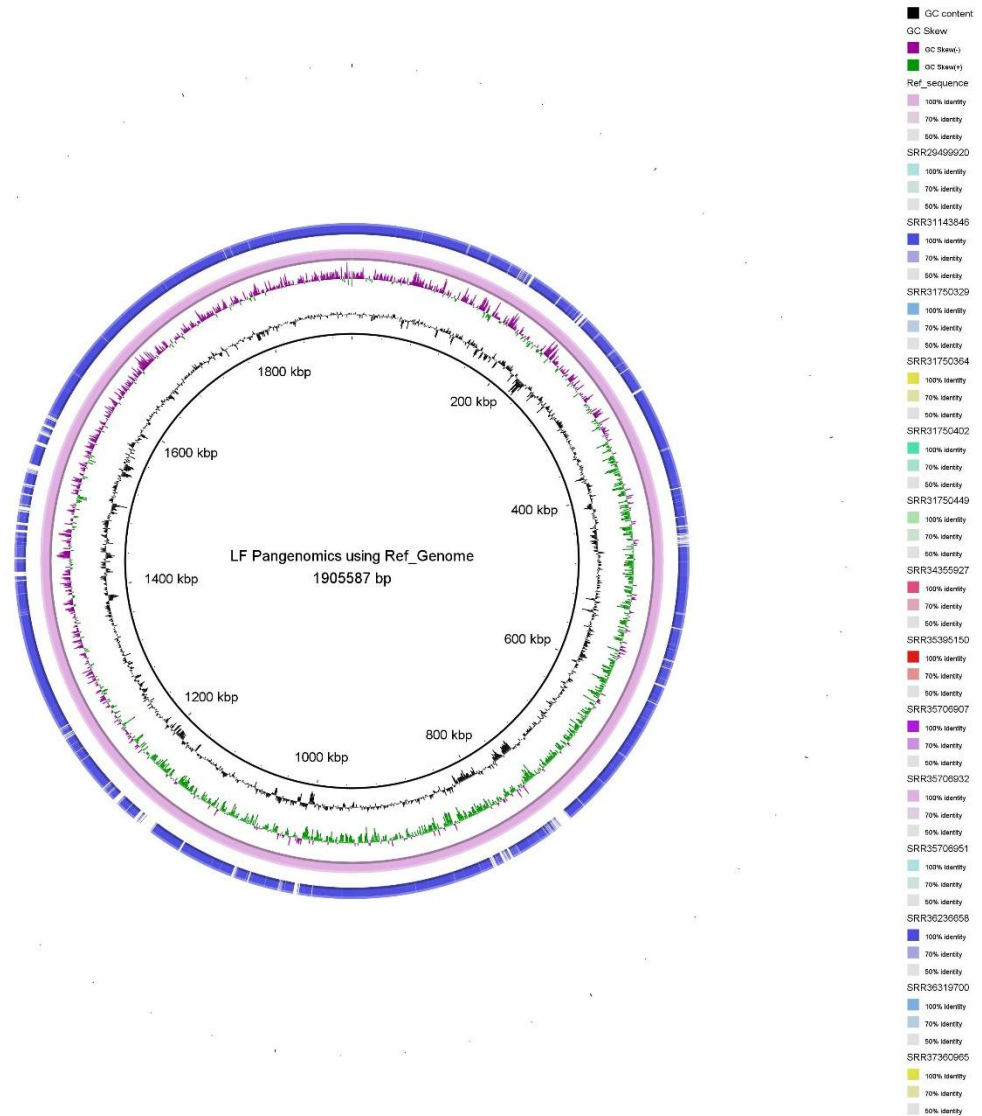


Figure 2. BRIG-based comparative genomic analysis of *Limosilactobacillus fermentum* isolates using a reference genome.

Circular visualization generated using BRIG comparing multiple *Limosilactobacillus fermentum* genomes against a reference genome of 1,905,587 bp. The concentric rings represent individual isolates aligned to the reference sequence. Color intensity corresponds to BLAST sequence identity thresholds of 100%, 70%, and 50%. Continuous colored regions indicate conserved genomic segments, whereas white gaps represent regions of reduced similarity or absence relative to the reference genome. The inner tracks show GC content (black) and GC skew

(magenta and green). The extensive continuity of the rings demonstrates high genomic conservation among isolates, while localized discontinuities identify strain-specific variable regions and potential accessory genomic elements.